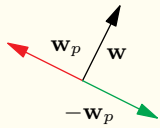
1.1.1

Gitt en vektor $\mathbf{w}$. Vi definerer $\mathbf{w}_p$ som vektoren som

- står vinkelrett på $\mathbf{w}$.
- peker til venstre med hensyn til retningen til $\mathbf{w}$.
- har lik lengde som $\mathbf{w}$.

Hvis $\mathbf{w} = [x, y]$, kan det vises at $\mathbf{w}_p = [-y, x]$, og at $-\mathbf{w}_p$ peker til høyre for $\mathbf{w}$.



Gitt vektorene $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$ og $\mathbf{v}_p = [-d, c]$. Vis at

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Svar

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

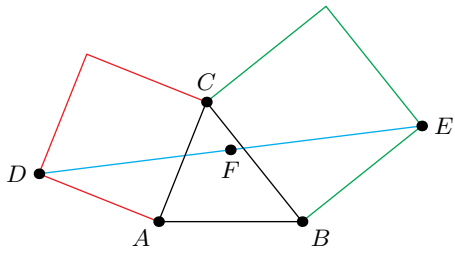
c)
$$\begin{aligned}(\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\ &= [-(b + d), a + c] \\ &= [-b, a] + [-d, c] \\ &= \mathbf{u}_p + \mathbf{v}_p\end{aligned}$$

1.1.2

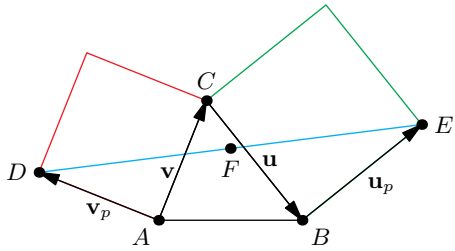
Den røde og den grønne firkanten er begge et kvadrat. Vi setter $\mathbf{v} = \overrightarrow{AC}$ og $\mathbf{u} = \overrightarrow{CB}$. F er midtpunktet på DE.

a) Finn koordinatene til F uttrykt ved $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$ og $\mathbf{v}_p$.

b) Beskriv den eventuelle forflytningen til F hvis C forskyves i en vilkårlig retning. (Det holder å se på tilfellet hvor C ligger over AB).



Svar



Vi innfører et koordinatsystem med origo i A. Da er $D = \mathbf{v}_p$. Videre er $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Videre er $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$ Dermed er

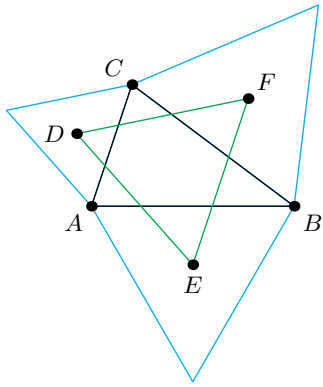
$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

$\mathbf{u}+\mathbf{v}$ og $\mathbf{u}_p+\mathbf{v}_p$ er ortogonale og har lik lengde. Så lenge $\mathbf{u}+\mathbf{v}$ ikke endres, vil altså ikke F forflyttes. En forskyvning av C medfører ingen endring av $\mathbf{u} + \mathbf{v}$, og dermed er poisjonen til F uavhengig av posisjonen til C.

1.1.3

De blå linjestykkene inngår i likesidede trekanter på utsiden av $\triangle ABC$. D, E og F er de respektive midtpunktene til de likesidede trekantene.

Vis at $\triangle DEF$ er likesidet.



.

Svar

Vi setter $\overrightarrow{AC} = 2\mathbf{v}$ og $\overrightarrow{AB} = 2\mathbf{u}$. Da er $\overrightarrow{BC} = 2\mathbf{v} - 2\mathbf{u}$. Videre har vi at

$$D = \mathbf{v} + k\mathbf{v}_p$$

$$E = \mathbf{u} - k\mathbf{u}_p$$

$$F = \mathbf{u} + \mathbf{v} - k(\mathbf{v}_p - \mathbf{u}_p)$$

hvor $k = \frac{1}{\sqrt{3}}$. Altså er

$$\overrightarrow{DF} = \mathbf{u} + k(\mathbf{u}_p - 2\mathbf{v}_p)$$
$$\overrightarrow{DE} = \mathbf{u} - \mathbf{v} - k(\mathbf{u}_p + \mathbf{v}_p)$$
$$\overrightarrow{EF} = \mathbf{v} + k(2\mathbf{u}_p - \mathbf{v}_p)$$

Da er

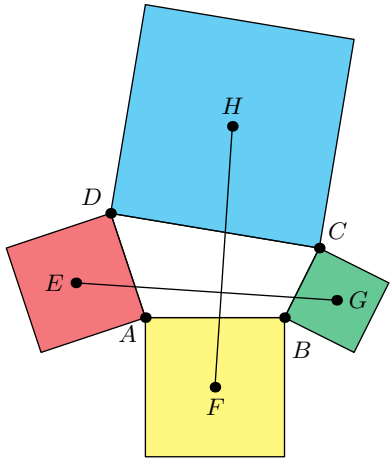
$$\begin{aligned} DF^2 &= \mathbf{u}^2 + 2k\mathbf{u}(\mathbf{u}_p - 2\mathbf{v}_p) + k^2(\mathbf{u}_p - 2\mathbf{v}_p)^2 \\ &= \mathbf{u}^2 - 4k\mathbf{u} \cdot \mathbf{v}_p + k^2(\mathbf{u}_p^2 - 4\mathbf{u}_p \cdot \mathbf{v}_p + 4\mathbf{v}_p^2) \\ &= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p \end{aligned}$$

$$
\begin{aligned}
DE^2 &= (\mathbf{u}-\mathbf{v})^2 - 2k(\mathbf{u}-\mathbf{v})(\mathbf{u}_p+\mathbf{v}_p) + k^2(\mathbf{u}_p+\mathbf{v}_p)^2 \\
&= \mathbf{u}^2 - 2\mathbf{u}\cdot\mathbf{v} + \mathbf{v}^2 - 2k(\mathbf{u}\cdot\mathbf{v}_p - \mathbf{u}_p\cdot\mathbf{v}) + k^2(\mathbf{u}_p^2 + 2\mathbf{u}_p\cdot\mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p\cdot\mathbf{v}_p) - 4k\mathbf{u}\cdot\mathbf{v}_p
\end{aligned}
$$

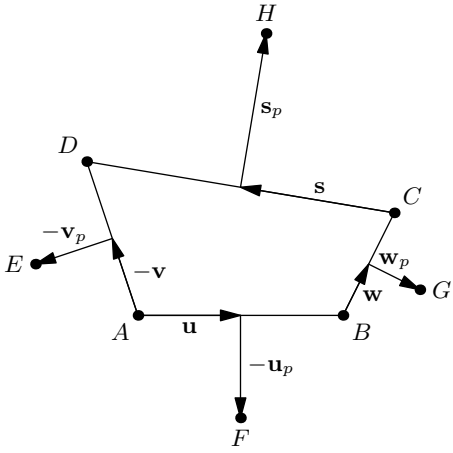
$$
\begin{aligned}
EF^2 &= \mathbf{v}^2 + 2k\mathbf{v}(2\mathbf{u}_p - \mathbf{v}_p) + k^2(2\mathbf{u}_p - \mathbf{v}_p)^2 \\
&= \mathbf{v}^2 + 4k\mathbf{u}_p\cdot\mathbf{v} + k^2(4\mathbf{u}_p - 4\mathbf{u}_p\mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p\cdot\mathbf{v}_p) - 4k\mathbf{u}\cdot\mathbf{v}_p
\end{aligned}
$$

1.1.4

De fargede firkantene er kvadrater. E, F, G og H er de respektive midtpunktene til kvadratene. Vis at $EG = FH$ og $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Svar



Vi innfører et koordinatsystem med origo i A, og setter $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, og $\overrightarrow{BC} = 2\mathbf{w}$ (i figuren er $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Da er

$$
\begin{aligned}
E &= -\mathbf{v} - \mathbf{v}_p \\
F &= \mathbf{u} - \mathbf{u}_p \\
G &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p \\
H &= \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Dermed er

$$
\begin{aligned}
\overrightarrow{EG} &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p \\
\overrightarrow{FH} &= \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Vi har at

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

Dette betyr at $\overrightarrow{FH}$ og $\overrightarrow{EG}$ er ortogonale. Videre er

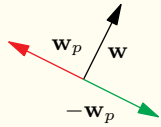
$$FH = |\overrightarrow{FH}_p| = | - \overrightarrow{EG}| = EG$$

1.1.5

Given a vector $\mathbf{w}$. We define $\mathbf{w}_p$ as the vector that

- is perpendicular to $\mathbf{w}$.
- points to the left with respect to the direction of $\mathbf{w}$.
- has the same length as $\mathbf{w}$.

If $\mathbf{w} = [x, y]$, it can be shown that $\mathbf{w}_p = [-y, x]$, and that $-\mathbf{w}_p$ points to the right of $\mathbf{w}$.



Given the vectors $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$, and $\mathbf{v}_p = [-d, c]$. Show that

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Answer

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

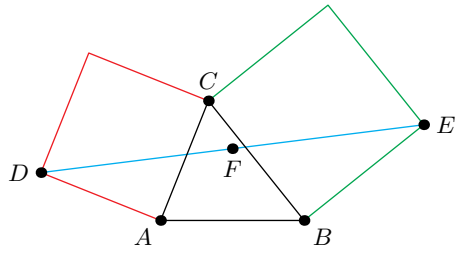
c)
$$\begin{aligned}(\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\ &= [-(b + d), a + c] \\ &= [-b, a] + [-d, c] \\ &= \mathbf{u}_p + \mathbf{v}_p\end{aligned}$$

Problem: Bottema's theorem

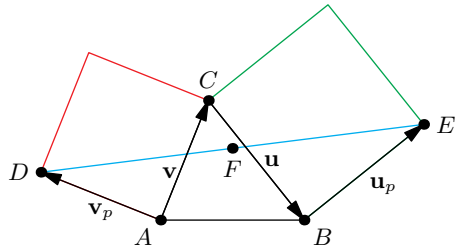
The red and the green quadrilateral are both squares. F is the midpoint of DE. Let $\mathbf{v} = \overrightarrow{AC}$ and $\mathbf{u} = \overrightarrow{CB}$. Given a vector $\mathbf{w}$, we define $\mathbf{w}_p$ as the vector perpendicular to $\mathbf{w}$, pointing to the left with respect to the direction of $\mathbf{w}$, and with equal length as $\mathbf{w}$.

a) Find the coordinates of F expressed in terms of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$, and $\mathbf{v}_p$.

b) Describe the possible displacement of F if C is shifted in an arbitrary direction. (It is sufficient to consider the case where C is above AB).



Solution



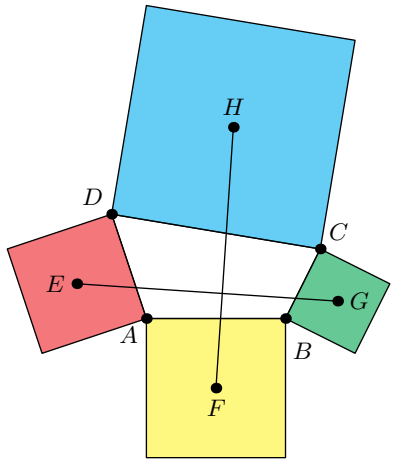
We introduce a coordinate system with the origin at A. Then $D = \mathbf{v}_p$. Furthermore, $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Furthermore, $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Thus,

$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

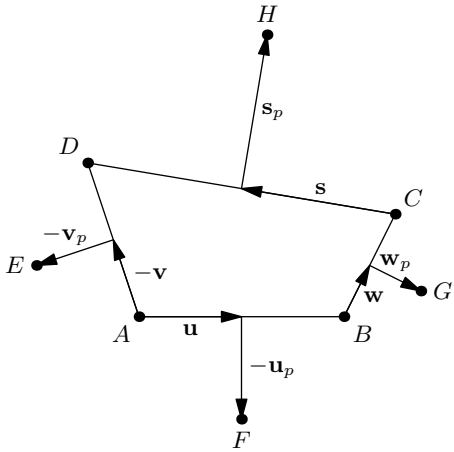
$\mathbf{u} + \mathbf{v}$ and $\mathbf{u}_p + \mathbf{v}_p$ are orthogonal and have the same length. As long as $\mathbf{u} + \mathbf{v}$ does not change, F will not be displaced. A displacement of C does not change $\mathbf{u} + \mathbf{v}$, and thus the position of F is independent of the position of C.

Problem: van Aubel's theorem

The colored quadrilaterals are squares. E, F, G, and H are the respective midpoints of the squares. Show that $EG = FH$ and $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Answer



We introduce a coordinate system with the origin at A, and set $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, and $\overrightarrow{BC} = 2\mathbf{w}$ (in the figure, $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Then,

$$
\begin{aligned}
E &= -\mathbf{v} - \mathbf{v}_p \\
F &= \mathbf{u} - \mathbf{u}_p \\
G &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p \\
H &= \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Thus,

$$\overrightarrow{EG} = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p$$
$$\overrightarrow{FH} = \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

We have

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

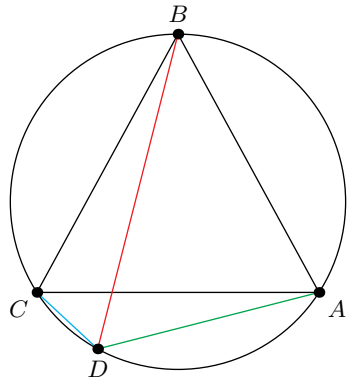
This means that $\overrightarrow{FH}$ and $\overrightarrow{EG}$ are orthogonal. Furthermore,

$$FH = |\overrightarrow{FH}_p| = | - \overrightarrow{EG}| = EG$$

A, B, C og D ligger på sirkelen, og $\angle ACB = \angle BAC = v$.

Vis at

$$BD = \frac{CD + DA}{2\cos v}$$



Svar

$\angle BDC$ og $\angle ADB$ spenner over samme bue som henholdsvis $\angle ACB$ og $\angle BAC$. Alle disse er periferivinkler, og da $\angle ACB = \angle BAC$, har de alle like størrelse. Videre følger det at $BC = AB$. Av cosinussetingen har vi da at

$$\begin{aligned} CD^2 + BD^2 - 2CD \cdot BD\cos v &= DA^2 + BD^2 - 2DA \cdot BD\cos v \\ CD^2 - DA^2 &= 2BD\cos v(CD - DA) \\ (CD + DA)(CD - DA) &= 2BD\cos v(CD - DA) \end{aligned}$$

Antatt at $CD - DA \neq 0$, er

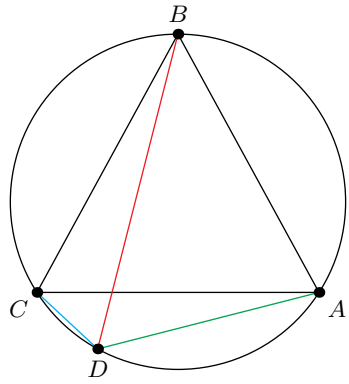
$$BD = \frac{CD + DA}{2\cos v}$$

I tilfellet hvor $CD = DA$, er BD en diameter i sirkelen. Da har vi av Thales' setning omvendt at $\angle DCB = 90°$. Altså er

$$BC = \frac{CD}{\cos v} = \frac{CD + DA}{2\cos v}$$

A, B, C and D lies on the circle, and $\angle ACB = \angle BAC = v$. Show that

$$BD = \frac{CD + DA}{2\cos v}$$

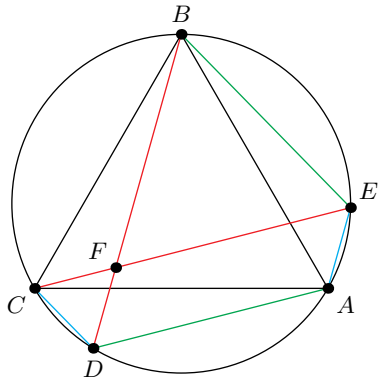


$\triangle ABC$ er likesidet, og $\triangle ACD \cong \triangle BAE$.

a) Avgjør hvilke av disse vinklene som har lik verdi. Bestem vinkelverdien om det lar seg gjøre.

$$\angle BDC, \angle ADB, \angle CFD, \angle ACE, \angle BEC, \angle EFB \text{ og } \angle CAD$$

b) Vis at $BD = CD + DA$.



Svar

a) $\angle BDC = \angle BEC = \angle BAC = 60°$ fordi de er periferivinkler som spenner over samme bue. Tlsvarende kan vi finne at $\angle ADB = 60°$. $\angle CAD = \angle ACE$ fordi de er periferivinkler som spenner over sirkelsegmenter med lik lengde. Av vinkelsummen til $\angle ADC$ er $\angle DCA = 60 - \angle CAD$. Dermed er

$$\angle DCF = \angle DCA + ACE = 60°$$

Altså er $\triangle CDF$ likesidet. $\angle EFB = \angle CFD$ fordi de er toppvinkler, og dermed er også $\triangle FEB$ likesidet. Følgelig har vi at

$$\angle BDC = \angle ADB = \angle CFD = \angle BEC = \angle EFB = 60°$$

og

$$\angle CAD = \angle ACE$$

b) Da $\triangle CDF$ og $\triangle FEB$ er likesidet, har vi at

$$BD = BF + FD = BE + CD = AD + CD$$

The Quatratic Function

Problem 1

Given the function $f(x) = ax^2 + bx + c$. Show that the graph of f is symmetric about the line $x = -\frac{b}{2a}$.

Answer

For the graph of f to be symmetric about the line $x = -\frac{b}{2a}$, for any number k, we must have

$$f\left(k - \frac{b}{2a}\right) = f\left(-k - \frac{b}{2a}\right) \tag{1.1}$$

For a number d, we have

$$d - \frac{b}{2a} = \frac{2ad - b}{2a}$$

Furthermore,

$$\begin{aligned} f\left(d - \frac{b}{2a}\right) &= a\left(\frac{2ad - b}{2a}\right)^2 + b\left(\frac{2ad - b}{2a}\right) + c \\ &= \frac{4a^2d^2 - 4abd + b^2}{4a} + \frac{2abd - b^2}{2a} + c \\ &= \frac{4a^2d^2 - b^2}{4a} + c \end{aligned}$$

This means that regardless of whether $d = k$ or $d = -k$, the expression above will be the same, and thus (1.1) holds.

Problem 2

Given the function $f(x) = ax^2 + bx + c$. In Problem 1, we showed that f is symmetric about the line $x = \frac{-b}{2a}$. Furthermore, it can be shown that for all numbers k,

$$f\left(\pm k - \frac{b}{2a}\right) = \frac{4a^2k^2 - b^2}{4a} + c$$

Use this to derive the *abc*-formula.

Answer

If $f(x) = ax^2 + bx + c = 0$, then for a number k, we have

$$f\left(\pm k - \frac{b}{2a}\right) = \frac{4a^2k^2 - b^2}{4a} + c = 0.$$

Solving this equation for k, we obtain

$$k = \pm\frac{\sqrt{b^2 - 4ac}}{2a}.$$

This means that

$$f\left(\pm\frac{\sqrt{b^2 - 4ac}}{2a} - \frac{b}{2a}\right) = 0.$$

Problem 3

Given the function $f(x) = ax^2 + bx + c$.

a) Find an expression for the extremal value.

b) Explain why

- a change in c will result in a vertical shift of the graph of f.
- a change in a or b will result in both a horizontal and a vertical shift of the graph of f.

Answer

In this solution, we assume the function has a minimum point. The case where it has a maximum point is entirely analogous.

a) Since $x = -\frac{b}{2a}$ is the extremal point of f, the extremal value f_e is given by

$$f_{\mathrm{e}} = f\left(-\frac{b}{2a}\right) = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c = -\frac{b^2}{4a} + c$$

b) Since the extremal point is expressed in terms of a and b, a change in c will not affect the horizontal position of the minimum point. However, c is part of the expression for f_e, so a change in c will alter the vertical position of the minimum point.

Both a and b are part of the expressions for the extremal point and f_e. Thus, a change in a or b will shift the minimum point both horizontally and vertically.

A shift in the minimum point necessarily corresponds to a shift in the graph of f.

Problem 4

Given the function $f(x) = ax^2 + bx + c$, where[1] $a > 0$. To describe the shape of the graph of f, we do the following[2]:

- Let $g(x)$ be the function that describes the difference between $f(x)$ and the minimum value of f.
- Let $h(g)$ be the horizontal distance between $g(x)$ and the line $x = -\frac{b}{2a}$.

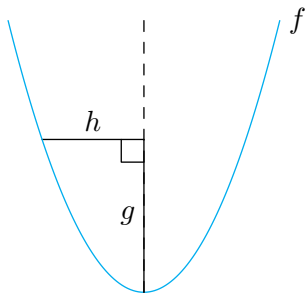
If two quadratic functions have identical expressions for $h(g)$, we can say that the functions have the same shape.

a) Show that

$$h(g) = \sqrt{\frac{g}{a}}$$

b) Refer back to Problem 3, and create a summary of how a change in a, b, or c might affect

- the horizontal shift of the graph,
- the vertical shift of the graph,
- the shape of the graph.



[1] The case where $a < 0$ is entirely analogous.

[2] The case where f has a maximum point instead of a minimum point is entirely identical.

Answer

a) In Problem 3, we found that the extremal value f_e of f is

$$f_e = -\frac{b^2}{4a} + c$$

Therefore,

$$g(x) = f(x) - f_e = ax^2 + bx + c - \left(-\frac{b^2}{4a} + c\right) = ax^2 + bx + \frac{b^2}{4a}$$

Solving this equation for x, we get from the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4a\left(\frac{b^2}{4a} - g\right)}}{2a} = \frac{-b \pm 2\sqrt{ag}}{2a}$$

Let $p(g) = x$ be the inverse function of g for $x \in [-\infty, -\frac{b}{2a}]$. Then,

$$p(g) = \frac{-b - 2\sqrt{ag}}{2a}$$

Furthermore,

$$h(g) = p(g) - \left(-\frac{b}{2a}\right) = \sqrt{\frac{g}{a}}$$

b) A change in c vertically shifts the graph but does not change its shape.

A change in b shifts the graph both horizontally and vertically but does not change its shape.

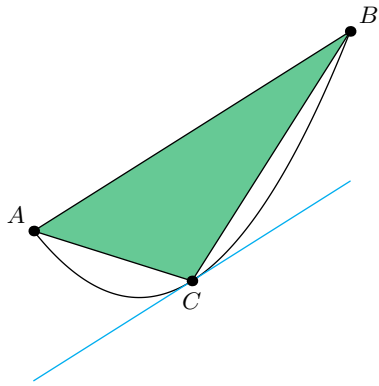
A change in a shifts the graph both horizontally and vertically and changes the shape of the graph.

Problem 5

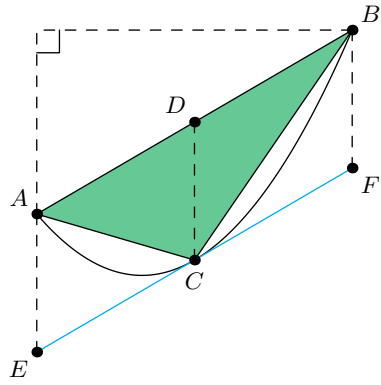
Given the function $f(x) = ax^2+bx+c$, and the points $A = (p, f(p))$ and $B = (q, f(q))$. The tangent to f at the point $C = (t, f(t))$ is parallel to the segment AB.

a) Show that $t = \frac{1}{2}(p + q)$.

b) Show that the area of $\triangle ABC$ is $\frac{1}{8}(q - p)^3$.



Answer



a) We have

$$A = (p, ap^2 + bp + c) \qquad \text{and} \qquad B = (q, aq^2 + bq + c)$$

Let $g(x)$ be the function describing the line through AB. Then,

$$\begin{aligned} g'(x) &= \frac{f(q) - f(p)}{q - p} \\ &= \frac{aq^2 + bq + c - (ap^2 + bp + c)}{q - p} \\ &= \frac{a(q^2 - p^2) + b(q - p)}{q - p} \\ &= \frac{a(q + p)(q - p) + b(q - p)}{q - p} \\ &= a(q + p) + b \end{aligned}$$

Let $C = (t, f(t))$. Then we have

$$\begin{aligned} f'(t) &= a(q + p) + b \\ 2at + b &= a(q + p) + b \\ t &= \frac{1}{2}(q + p) \end{aligned}$$

b) Let D be the midpoint of AB. Then,

$$D = \left(t, \frac{1}{2}(f(q) + f(p))\right)$$

Furthermore,

$$\begin{aligned}
DC &= \frac{1}{2}\left(f(q)+f(p)\right)-f(t)\\
&= \frac{1}{2}\Big(a(p^2+q^2)+b(p+q)+2c\Big)-a\left(\frac{1}{2}(p+q)\right)^2+b\left(\frac{1}{2}(p+q)\right)+c\\
&= \frac{a}{2}(p^2+q^2)-\frac{a}{4}(p+q)^2\\
&= \frac{a}{4}(2p^2+2q^2-p^2-q^2-2pq)\\
&= \frac{a}{4}(q-p)^2
\end{aligned}$$

We place E and F such that $\square AEFB$ is a parallelogram. With AE as the base, the height of $\square AEFB$ is $q-p$. Since $AE=DC$, we have

$$A_{\triangle ABC}=\frac{1}{2}A_{\square AEFB}=\frac{1}{2}\cdot\frac{a}{4}(q-p)^2(q-p)=\frac{a}{8}(q-p)^3$$

Problem 6

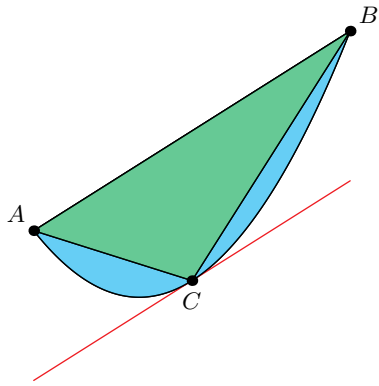
Given the function $f(x) = ax^2+bx+c$ and the points $A = (p, f(p))$ og $B = (q, f(q))$.

a) Let I be the area bounded by the graph of f and the line segment AB. Determine which of a, b, c, p and q we can expect to be included in the expression for I.

b) Find the expression for I.

c) Find $\frac{I}{A_{\triangle ABC}}$.

d) Find $\frac{A_{\triangle ABC}}{A_{\text{blue area}}}$.



Answer

a) We expect p and q to appear in the expression for I since these determine the positions of A and B. As long as the shape of f remains the same, the bounded area will not change. Since only a affects the shape of f, a must also appear in the expression for I, but not b and c.

b) Let $g(x)$ be the function describing the line through AB. Then

$$\begin{aligned} g(x) &= (a(p+q)+b)(x-p)+f(p) \\ &= (a(p+q)+b)x-(a(p+q)+b)p+ap^2+bp+c \\ &= (a(p+q)+b)x-apq+c \end{aligned}$$

We have that

$$g-f = ax(p+q)-apq-ax^2$$

Therefore,

$$\begin{aligned} I &= \int_p^q (g-f)\,dx \\ &= \left[\frac{ax^2}{2}(p+q)-apqx-\frac{1}{3}ax^3\right]_p^q \\ &= \frac{aq^2}{2}(p+q)-apq^2-\frac{1}{3}aq^3-\left(\frac{ap^2}{2}(p+q)-ap^2q-\frac{1}{3}ap^3\right) \\ &= \frac{a}{2}(p+q)(q^2-p^2)-apq(q-p)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)((p+q)^2-2pq)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)(p^2+q^2)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)(p^2+q^2)-\frac{a}{3}(q-p)(p^2+q^2+pq) \\ &= \frac{a}{6}(q-p)(3p^2+q^2-2p^2-2q^2-2pq) \\ &= \frac{a}{6}(q-p)^3 \end{aligned}$$

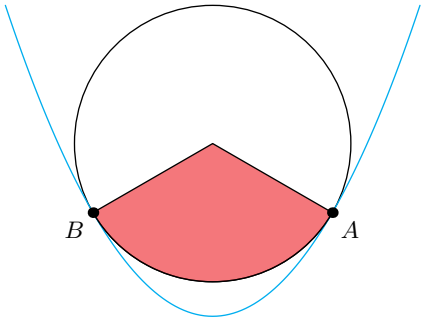
c) Since $I = \frac{a}{6}(q-p)^3$ and $A_{\triangle ABC} = \frac{a}{8}(q-p)^3$, we have

$$\frac{I}{A_{\triangle ABC}} = \frac{4}{3}$$

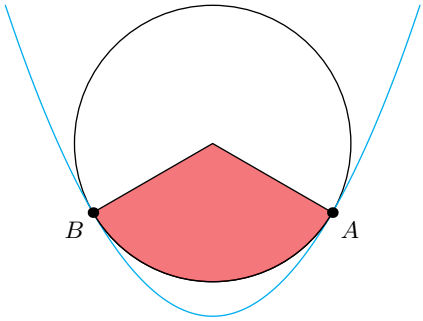
d) We have that

$$A_{\text{blue area}} + A_{\triangle ABC} = \frac{4}{3} A_{\triangle ABC}$$
$$\frac{A_{\triangle ABC}}{A_{\text{blue area}}} = 3$$

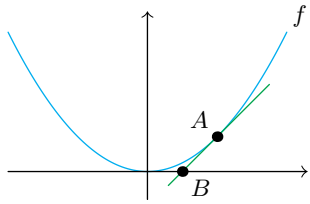
Grafen til $f = x^2 + \frac{3}{4}$ tangerer sirkelen i A og B. $AB = \sqrt{3}$.
Finn arealet til den røde sektoren.



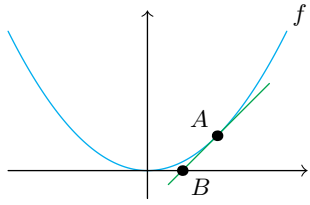
The graph of $f = x^2 + \frac{3}{4}$ are tangent to the circle at A and B.
$AB = \sqrt{3}$.
Find the area of the red sector.



Gitt funksjonen $f(x) = ax^2$. Tangenten til f i punktet A skjærer x-aksen i punktet B. Vis at hvis $A = (c, f'(c))$, så er $B = \left(\frac{c}{2}, 0\right)$.



Given the function $f(x) = ax^2$. The tangent to f at point A intersects the x-axis at point B. Show that if $A = (c, f(c))$, then $B = \left(\frac{c}{2}, 0\right)$.



Gitt funksjonen $f(x) = x^2$. Det finnes et **brennpunkt** $F = (0, p)$ og en **styrelinje** gitt ved $s(x) = -p$ slik at $AF = AB$, hvor A ligger på grafen til f, $AF \perp AB$, og $p \neq 0$.

a) Finn verdien til p.

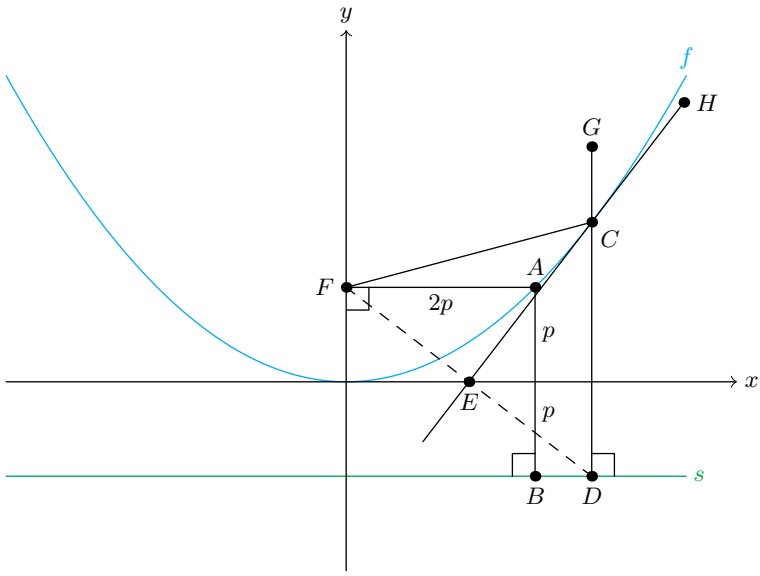
Vi setter $C = (c, f(c))$.

b) Finn lengden til FC uttrykt ved c.

c) Vis at $FC = CD$.

d) E er skjæringspunktet til tangenten til f i C og x-aksen. Finn x-koordinaten til E, og vis at E ligger på FD.

e) Vis at $\angle HCG = \angle FCE$.



Given the function $f(x) = x^2$. There exists a **focus** $F = (0, p)$ and a **directrix** given by $s(x) = -p$ such that $AF = AB$, where A lies on the graph of f, $AF \perp AB$, and $p \neq 0$.

a) Find the value of p.

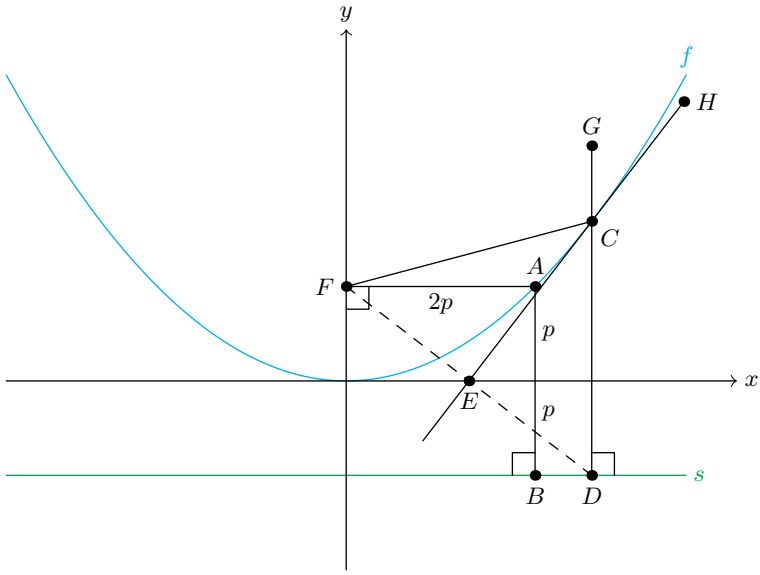
Let $C = (c, f(c))$.

b) Find the length of FC expressed in terms of c.

c) Show that $FC = CD$.

d) E is the intersection point of the tangent to f at C and the x-axis. Find the x-coordinate of E, and show that E lies on FD.

e) Show that $\angle HCG = \angle FCE$.



Given a logistic function

Given a logistic function

$$f(x) = \frac{a}{1 + Ce^{-bx}}$$

where $a, b, C > 0$.

a) Show that $f' = \frac{b}{a} f \cdot (a - f)$.

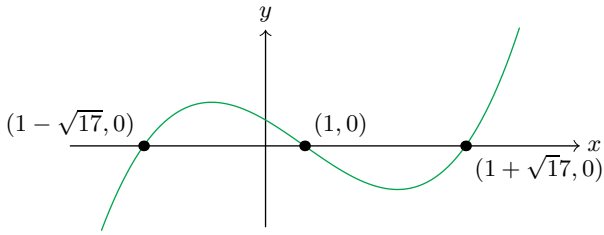
b) Let d be the number such that $f''(d) = 0$. Find d.

c) Show that $f(d + x) + f(d - x) = a$.

d) Let $h_0 = f(d) - f(d - x)$ and $h_1 = f(d + x) - f(d)$. Show that $h_0 = h_1$. Provide a geometric interpretation of this identity.

e) Show that $f''(d + x) = -f''(d - x)$. What point does this identity suggest that $x = d$ is for f? What point is $(d, f(d))$ for f?

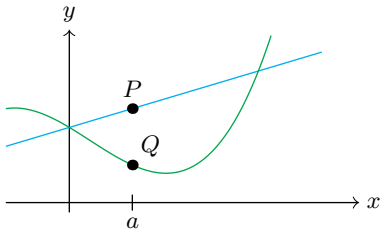
Gruble* 1 (1TV23)
Figuren under viser grafen til den deriverte funksjonen til $f(x)$. Null-punktene til f er $x = -4$, $x = -2$, $x = 4$ og $x = 6$. Lag en skisse som viser hvordan grafen til f kan se ut. Grunngi svaret ditt.



Gruble* 2 (1TH23)
Figuren under viser de respektive grafene til funksjonene $f(x) = 2x + 8$ og $g(x) = x^3 - x^2 - 4x + 8$. $P = (a, f(a))$ og $Q = (a, g(a))$.

a) Finn lengden til QP når $a = 1$.

b) Finn den største lengden til QP når $a \in (0, 3)$



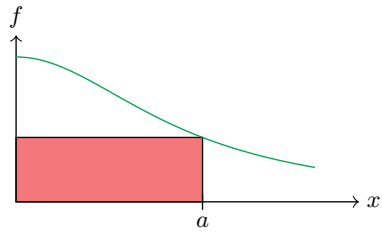
Gruble* 3 (1TH23)
Figuren under viser grafen til funksjonen

$$\frac{8}{x^2+20} \quad , \quad x > 0$$

og et rektangel med hjørner $(0,0)$, $(a,0)$, $(a,f(a))$ og $(0,f(a))$.

a) Finn arealet til rektangelet når $a = 5$.

b) Finn a slik at arealet til rektangelet blir så stort som mulig.



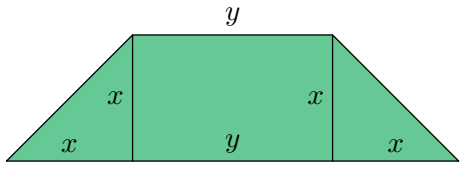
Gruble* 4 (1TH24)
Figuren under består av to rettvinklede trekanter og et rektangel. Den samlede lengden av alle linjestykkene med markerte lengder og de to hypotenusene er 100.

a) Finn arealet til figuren når $x = 8$.

b) Finn y uttrykt ved x.

c) Finn arealet A til figuren uttrykt ved x.

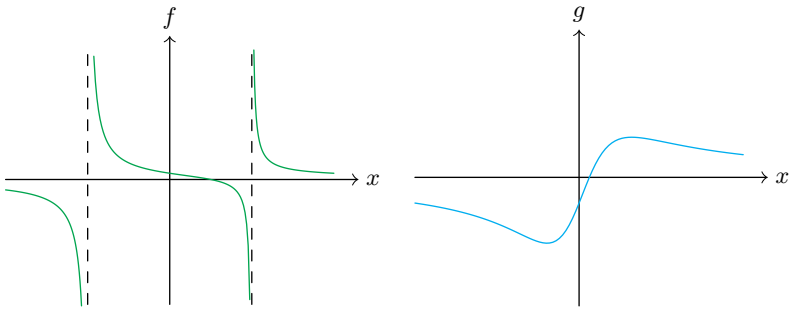
d) Finn x slik at A blir størst mulig.



Gruble* 5 (1TV25)

Under vises de respektive grafene til de rasjonale funksjonene $f(x)$ og $g(x)$. Avgjør hvilke av påstenden som passer til hvilken funksjon. Stiplede linjer viser vertikale asymptoter.

- For én verdi av x går teller mot en verdi forskjellig fra 0, mens nevner går mot 0.
- Nevner går aldri mot null, og teller er en konstant.
- For to verdier av x går teller mot en verdi forskjellig fra 0, mens nevner går mot 0.
- Nevner går aldri mot null, og teller er ikke en konstant.



Gruble 6

Gitt funksjonen

$$f(x) = \frac{1}{6}x(1350 - 136\sqrt{x} + 3x) \quad , \quad x > 0$$

Finn toppunktet og bunnpunktet til f.